Reduction Formulae for Integration

Questions 1–30

Question 1. If $I_n = \int x^n e^x \, dx$, show that for $n \geq 1$,

$$I_n = x^n e^x - nI_{n-1}.$$

Question 2. For $I_n = \int x^n e^x \, dx$ it is given that $I_n = x^n e^x - nI_{n-1}$ for $n \geq 1$. Evaluate each of the following:

a) $\int x^2 e^x \, dx$

b) $\int x^3 e^x \, dx$

c) $\int x^4 e^x \, dx$

Question 3. If $I_n = \int x^n e^{2x}\,dx$, show that for $n \geq 1$,

$$I_n = \frac{1}{2}x^n e^{2x} - \frac{n}{2}I_{n-1}.$$

Question 4. For $I_n = \int x^n e^{2x}\,dx$ it is given that $I_n = \frac{1}{2}x^n e^{2x} - \frac{n}{2}I_{n-1}$ for $n \geq 1$. Evaluate each of the following:

a) $\int x^3 e^{2x}\,dx$

b) $\int x^4 e^{2x}\,dx$

Question 5. If $I_n = \int x(\ln x)^n \, dx$, show that for $n \geq 2$,

$$I_n = \frac{x^2}{2}(\ln x)^n - \frac{n}{2}I_{n-1}.$$

Question 6. For $I_n = \int x(\ln x)^n \, dx$ it is given that $I_n = \frac{x^2}{2}(\ln x)^n - \frac{n}{2}I_{n-1}$ for $n \geq 2$. Evaluate each of the following:

a) $\int x(\ln x)^2 \, dx$

b) $\int x(\ln x)^3 \, dx$

Question 7. If $I_n = \int \sin^n x \, dx$, show that for $n \geq 2$,

$$I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}.$$

Question 8. For $I_n = \int \sin^n x \, dx$ it is given that $I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$ for $n \geq 2$. Evaluate each of the following:

a) $\int \sin^4 x \, dx$

b) $\int \sin^5 x \, dx$

Question 9. If $I_n = \int \cos^n x \, dx$, show that for $n \geq 2$,

$$I_n = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}.$$

Question 10. For $I_n = \int \cos^n x \, dx$ it is given that $I_n = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$ for $n \geq 2$. Evaluate each of the following:

a) $\int \cos^4 x \, dx$

b) $\int \cos^5 x \, dx$

Prerequisite Knowledge for Questions 11–12: Calculus of Hyperbolic Functions

Question 11. Let $I_n = \int \sinh^n x \, dx$.

a) Show that for $n \geq 2$, $I_n = \dfrac{\sinh^{n-1} x \cosh x}{n} - \dfrac{n-1}{n} I_{n-2}$.

b) Hence evaluate $\int \sinh^4 x \, dx$.

Question 12. Let $I_n = \int \cosh^n x \, dx$.

a) Show that for $n \geq 2$, $I_n = \dfrac{\cosh^{n-1} x \sinh x}{n} + \dfrac{n-1}{n} I_{n-2}$.

b) Hence evaluate $\int \cosh^6 x \, dx$.

Question 13. Let $I_n = \int x^n \sqrt{1+x}\, dx$.

a) Show that for $n \geq 1$,

$$I_n = \frac{2x^n(1+x)^{3/2}}{2n+3} - \frac{2n}{2n+3} I_{n-1}.$$

b) Hence evaluate $\int x^2 \sqrt{1+x}\, dx$.

Question 14. Let $I_n = \int \tan^n x\, dx$.

a) By considering $1 + \tan^2 x = \sec^2 x$, show that for $n \geq 2$,

$$I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}.$$

b) Show that $\int \tan x\, dx = \ln|\sec x| + c$.

c) Hence evaluate $\int \tan^5 x\, dx$.

Question 15. Let $I_{m,n} = \int x^m (\ln x)^n \, dx$.

a) Show that for $n \geq 1$,

$$I_{m,n} = \frac{x^{m+1}(\ln x)^n}{m+1} - \frac{n}{m+1} I_{m,n-1}.$$

b) Hence evaluate $\int x^3 \ln x \, dx$.

Definite Integrals

Question 16. Let $I_n = \int_0^1 \frac{x^n}{\sqrt{1-x}} \, dx$.

a) Show that for $n \geq 1$, $I_n = \frac{2n}{1+2n} I_{n-1}$.

b) Hence show that $\int_0^1 \frac{x^4}{\sqrt{1-x}} \, dx = \frac{256}{315}$.

Question 17. Let $I_n = \int_0^{\pi/2} x^n \sin x \, dx$.

a) Show that for $n \geq 2$, $I_n = n\left(\frac{\pi}{2}\right)^{n-1} - n(n-1)I_{n-2}$.

b) Hence show that $\int_0^{\pi/2} x^4 \sin x \, dx = \frac{1}{2}(48 - 24\pi + \pi^3)$.

Question 18. Let $I_n = \int_0^{2\pi} \cos^n x \, dx$. By considering the reduction formula in each case, show that when:

a) n is odd, $I_n = 0$

b) n is even, $I_n = \left(\frac{2n-1}{2n}\right)\left(\frac{2n-3}{2n-2}\right)\cdots\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)2\pi$.

Question 19. Let $I_n = \int_0^{2\pi} \sin^n x \, dx$. By considering the reduction formula in each case, show that when:

a) n is odd, $I_n = 0$

b) n is even, $I_n = \left(\frac{2n-1}{2n}\right)\left(\frac{2n-3}{2n-2}\right)\cdots\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)2\pi$.

Question 20. Let $I_n = \int_0^{\pi/3} \sec^n x \, dx$.

a) Show that for $n \geq 1$, $(n-1)I_n = 2^{n-2}\sqrt{3} + (n-2)I_{n-2}$.

b) Hence evaluate $\int_0^{\pi/3} \sec^8 x \, dx$.

Question 21. Let $I_n = \int_0^{\pi/3} \sin^n x \, dx$.

a) Show that for $n \geq 2$, $I_n = \frac{n-1}{n} I_{n-2} - \frac{1}{2n}\left(\frac{\sqrt{3}}{2}\right)^{n-1}$.

b) Hence show that $\int_0^{\pi/3} \sin^6 x \, dx = a\pi - b\sqrt{3}$, where a and b are constants to be found.

c) State the value of $\int_{-\pi/3}^{\pi/3} \sin^6 x \, dx$.

Question 22. Let $I_n = \int_0^{\pi/2} \cos^n x \, dx$. Given that for $n \geq 2$, $I_n = \frac{n-1}{n} I_{n-2}$, evaluate $\int_{-\pi/2}^{\pi/2} \cos^6(3t) \, dt$.

Question 23. Let $I_n = \int_0^1 (1-x^2)^n \, dx$.

a) Show that for $n \geq 1$, $I_n = \dfrac{2n}{2n+1} I_{n-1}$.

b) Hence evaluate $\int_{-1}^1 (1-x^2)^6 \, dx$.

Question 24. Given that $n \geq 2$, if $I_n = \int_0^{\pi/2} \cos^n x \, dx$ then $I_n = \dfrac{n-1}{n} I_{n-2}$.

a) Show that for $n \geq 1$, $I_n = \dfrac{2n-1}{2n+2} I_{n-1}$.

b) Hence show that $\int_0^{\pi/2} \cos^{12} x \sin^2 x \, dx = \dfrac{33\pi}{4096}$.

c) State the value of $\int_{-\pi/2}^{\pi/2} \cos^{12} x \sin^2 x \, dx$.

Prerequisite Knowledge for Question 25: Calculus of Hyperbolic Functions

Question 25. Let $I_n = \displaystyle\int_0^{\ln 3} \tanh^n x \, dx$.

a) By considering $\tanh^2 x + \operatorname{sech}^2 x = 1$, show that for $n \geq 2$,

$$I_n = I_{n-2} - \frac{1}{n-1}\left(\frac{4}{5}\right)^{n-1}.$$

b) Hence show that $\displaystyle\int_0^{\ln 3} \tanh^4 x \, dx = \ln a - \frac{b}{c}$, where a, b and c are integers to be found.

Question 26. Let $I_n = \displaystyle\int_0^1 x^n e^{ax} \, dx$ for constant $a \neq 0$.

a) Show that for $n \geq 1$, $I_n = \dfrac{e^a}{a} - \dfrac{n}{a} I_{n-1}$.

b) Hence determine the value of $\displaystyle\int_0^1 x^2 e^{x/4} \, dx$.

Question 27. Let $I_n = \int_0^{\sqrt{a}} (a - x^2)^n \, dx$ where a is a positive constant.

a) Show that for $n \geq 1$, $I_n = \dfrac{2na}{2n+1} I_{n-1}$.

b) Hence evaluate $\int_0^4 (16 - x^2)^3 \, dx$.

Question 28. Let $I_n = \int_0^{\pi} \dfrac{\cos nx}{\cos x} \, dx$.

a) Given $\cos(A + B) + \cos(A - B) = 2\cos A \cos B$, show that

$$\frac{\cos nx}{\cos x} = 2\cos((n-1)x) - \frac{\cos((n-2)x)}{\cos x}.$$

b) Show that for $n \geq 2$, $I_n = -I_{n-2}$.

c) Hence show that $\int_0^{\pi} \dfrac{\cos 7x}{\cos x} \, dx = -\pi$.

d) Evaluate $\int_0^{\pi} \dfrac{\cos 53x}{\cos x} \, dx$.

Question 29. Let $I_{m,n} = \int_0^k x^n(a-x)^m\,dx$ for constant $k < a$.

a) Show that for $n \geq 1$,

$$I_{m,n} = -\frac{k^n(a-k)^{m+1}}{m+1} + \frac{n}{m+1}I_{m+1,n-1}.$$

b) Hence show that $\int_0^1 \frac{x^4}{(2-x)^3}\,dx = a\ln 2 - b$, where a and b are positive constants to be found.

Prerequisite Knowledge for Question 30: L'Hôpital's Rule and Improper Integrals

Question 30. Let $I_n = \int_0^1 (\ln x)^n\,dx$.

a) Find $\lim_{x\to 0^+}(x\ln x)$.

b) Show that for $n \geq 1$, $I_n = -nI_{n-1}$.

c) Hence show that for $n \geq 1$, $I_n = (-1)^n n!$.

d) Evaluate $\int_0^1 (\ln x)^4\,dx$.